MASTER OF BUSINESS ADMINISTRATION

SEMESTER 3

DADS304

VISUALIZATION

Unit 6

Introduction to Tableau

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1. INTRODUCTION

Tableau is a data visualization tool or business intelligence tool/software which analyzes and displays data in a chart or report easily. It is very easy to use because it does not require any coding skill.

Users can build and distribute an interactive and shareable dashboard, which shows the trends, variations, and density of the data in the form of graphs and charts and tables. Tableau can connect to files, relational and Big Data sources to acquire and preprocess data. The software allows combining data from multiple sources and real-time collaboration, which makes it unique. It is used by businesses, academic researchers, and many government organizations for visual data analysis. It is also positioned in top Business Intelligence and Analytics Platform in Gartner Magic Quadrant

Tableau offers numerous appealing and distinctive features that make it a top tool for data visualization. You can quickly get the answers to crucial queries thanks to its robust data finding and exploration application. Tableau's drag and drop interface makes it simple to explore various views, merge numerous databases, and visualize any type of data. It doesn't need any difficult scripting. Anyone who is familiar with the business issues can solve them by visualizing the pertinent facts. Sharing results with others is as simple as publishing to Tableau Server after analysis.

1.1 Learning Objectives

- *Data visualization basics.*
- *Tableau software and its applications in different industries.*

2. TABLEAU FEATURES

Tableau offers solutions for various departments, industries, and data settings. The following special attributes allow tableau to handle a variety of scenarios.

Speed of Analysis Since it does not call for a high level of programming knowledge, any user with access to data can begin utilizing it to extract value from the data.

- Tableau is self-sufficient in that it doesn't require a convoluted software setup. Most customers utilize the desktop version, which is simple to install and has all the functionality required to begin and finish data analysis.
- The user investigates and evaluates the data using visual tools including colors, trend lines, charts, and graphs. Almost everything can be done by drag and drop, thus there is very little script to write.
- Integrate Diverse Data Sets Tableau enables you to instantly combine many relational, semi-structured, and unstructured data sources without incurring high upfront integration expenses.
- Tableau operates on all types of devices where data flows, regardless of architecture. As a result, the user does not have to be concerned with specific hardware or software requirements to utilize Tableau.
- Real-Time Collaboration – Tableau can embed a live dashboard in portals like Salesforce or a SharePoint site and filter, sort, and debate data instantly. By simply refreshing their web browser, colleagues can see the most recent data by subscribing to your interactive dashboards and saving your perspective of the data.
- All the organization's published data sources may be managed in one place thanks to Tableau server. In one simple place, you can remove, modify permissions, add tags, and manage schedules. Extract refreshes may be easily scheduled and managed on the data server.

3. WHAT IS DATA VISUALIZATION?

Data visualization uses visual components like graphs, charts, and maps to graphically portray quantitative information and data. Data visualization turns both huge and small data sets into graphics that are simple for people to comprehend and process. Data outliers, patterns, and trends can be easily understood with data visualization tools. The tools and technology for data visualization are essential in the realm of big data because they allow for the analysis of enormous amounts of data.

Top Data Visualization Tools

Tableau: For visual analytics, there is a tableau desktop application. You can visualize your reports online and on mobile devices with a server option if you don't want to install tableau software on your desktop. For individuals who desire the server solution but don't want to set it up manually, a cloud-hosted service is also an option. Citrix, Pandora, and Barclays are some of Tableau's clients.

Infogram: With Infogram, even non-designers can produce powerful data visualizations for marketing reports, infographics, social media posts, maps, dashboards, and more. Infogram is a fully featured drag-and-drop visualization tool. The following file types can be used to export finished visualizations: PNG,.JPG,.GIF,.PDF, and.HTML. Additionally, interactive visualizations are feasible and ideal for integrating into websites and applications. Additionally, Infogram provides a WordPress plugin that streamlines the process of integrating visualizations for WordPress users.

Power BI: A business intelligence (BI) platform called Microsoft Power BI gives non-technical business people the means to gather, analyze, visualize, and share data. With its strong interaction with other Microsoft products, Power BI is a versatile self-service tool that requires little initial training. Its user interface is intuitive for Excel users.

Chartblocks: According to ChartBlocks, data may be loaded using their API from "everywhere," even live streams. Even though they claim that it only takes a few clicks to import data from any source, it is undoubtedly more difficult to use than other programs that have automatic modules or extensions for particular data sources. The final representation produced by the software can be heavily customized, and the chart construction wizard

assists users in selecting the ideal data for their charts before importing the data. A major benefit for data visualization designers who wish to embed charts into websites that are likely to be viewed on several devices is that designers may construct almost any type of chart, and the output is responsive.

Data wrapper: To include charts and maps in news reports, Data wrapper was developed. The produced maps and charts can be embedded on news websites and are interactive. However, they only have a few data sources, and the main approach is to copy and paste data into the program. Charts can be produced after data has been imported with only one click. They use a variety of visualizations, including choropleth and symbol maps, column, line, and bar charts, election donuts, area charts, scatter plots, and locator maps. The final visuals resemble those that may be found on websites like the New York Times or Boston Globe. In fact, magazines like Mother Jones, Fortune, and The Times use their charts.

Plotly: An interactive, open-source, and browser-based Python graphing toolkit is called plotly.py.Plotly.py is a high-level, declarative charting framework that is built on top of plotly.js. Plotly.js comes with more than 30 different chart kinds, including financial charts, scientific charts, 3D graphs, and more. Plotly is MIT licensed software. Plotly graphs can be seen in standalone HTML files, Jupyter notebooks, or Dash programmers' advice, dashboard creation, app integration, and feature requests, get in touch with us.

RAW: RAW, also referred to as RawGraphs, operates on delimited data such as TSV or CSV files. It acts as a bridge between spreadsheets and data visualization. Despite being a web-based application, RawGraphs offers strong data protection and offers a variety of unconventional and traditional layouts.

Data Visualization's Value

Because of how the human brain processes information, data visualization is crucial. It is more pleasant to study graphs and charts than spreadsheets and reports when visualizing vast amounts of complex data sets.

An efficient and rapid technique to communicate ideas to everyone is through data visualization. By making a small tweak, you can try out a new outline.

SELF-ASSESSMENT QUESTIONS – 1

1. Tableau is a coding language for software. True/False
2. Tableau was introduced in the year of _____.

4. HISTORY OF TABLEAU

Stanford University students Pat Hanrahan, Christian Chabot, and Chris Stolte launched Tableau in 2003. Making the database industry dynamic and comprehensive was the fundamental motivation for its inception. Tableau debuted at a time when Cognos, Microsoft Excel, and Business Objects were already well-known brands.

The main features that led Tableau Software to achieve success are:

- VizQL is the language that powers it, increasing the flexibility to pull data from any source.
- Provide the user with the ability to alter Tableau reports using a variety of visualization tools.
- The drag-and-drop method can be used to create any complex graphs or maps.
- Multiple platforms allow for the insertion of Tableau data visualizations.
- Real-time data analysis and visualization are both possible.

5. GETTING STARTED WITH TABLEAU

Installation Tableau Desktop

Download Tableau Desktop's Free Personal Edition from Tableau Desktop. To download, you must first register with your information. You must accept the licensing agreement and specify the installation target folder during the very simple installation process that follows downloading. The whole setup procedure is detailed in the stages and screenshots that follow.

1. Start installation and click on "run" when asked

Fig 1.1: Starting installation

2. Accept the License Agreement

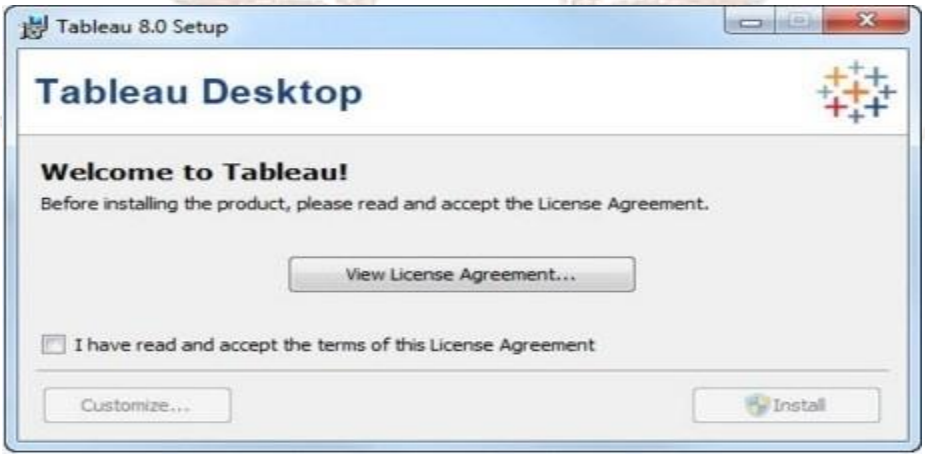

Fig 1.2: Accept license

3. Start Trial

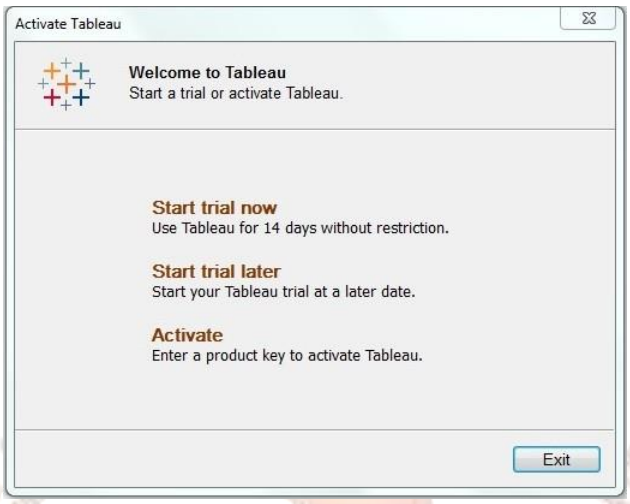

Fig 1.3: Start trial

4. Provide Your Details

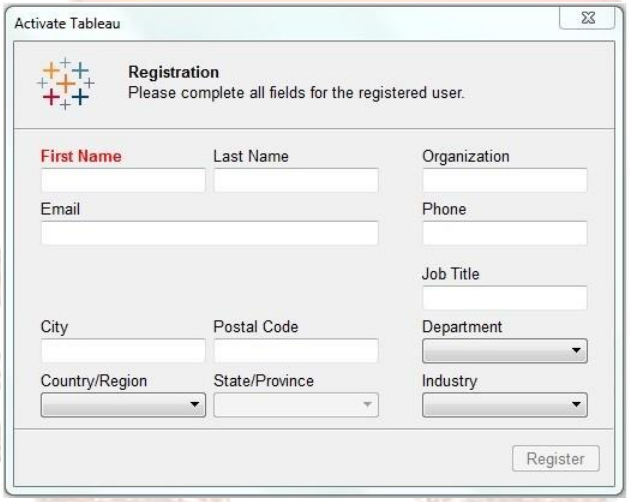

Fig 1.4: Details

5. Registration Complete

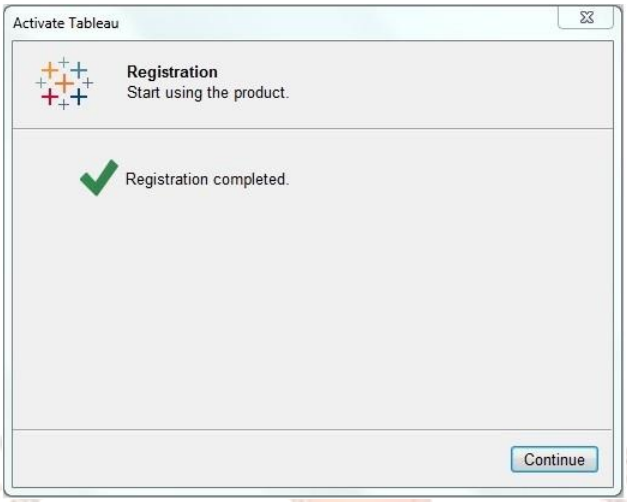

Fig 1.5: Installation done

6. Verify the Installation

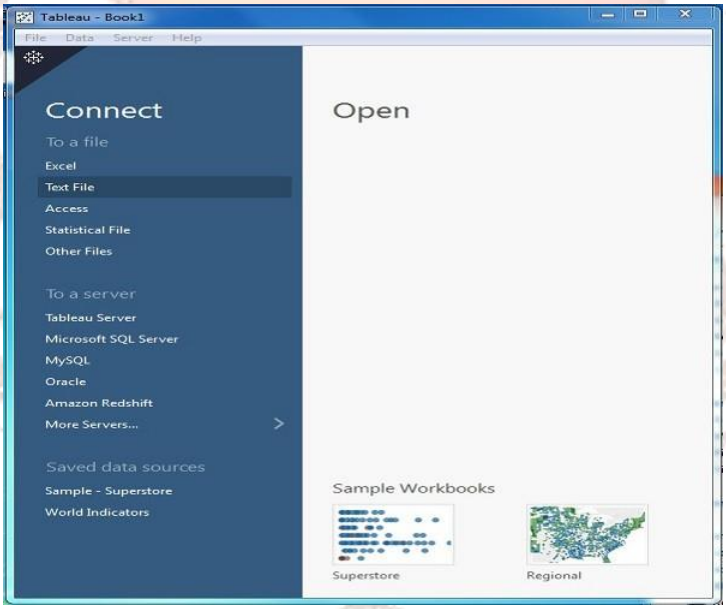

Fig 1.6: Verification

6. TABLEAU FILE TYPES

After data processing, Tableau's output can be saved in a variety of formats and then delivered across several platforms. The several distinct extensions serve to distinguish the various types of different file categories. Their length varies depending on how they are produced and how they are used.

Type	File Extension	Purpose
Tableau workbook	(.twb)	Tableau workbook can hold one or more worksheets, and also hold zero or more stories and dashboards.
Tableau Bookmarks	(.tbm)	Tableau bookmarks can hold a single worksheet that can be easily shared, and pasted into other workbooks.
Tableau Packaged workbook	(.twbx)	Tableau packaged workbook is a single zip file which contains a workbook along with any supporting local file data and background images. This is the best way to package your work for sharing with others who don't have access to the original data.
Tableau data Extract	(.hyper or .tde)	Tableau data extract is a local copy of the entire data set. It is used to share the data with others when you worked offline, and want to improve the performance.
Tableau data Source	(.tds)	Tableau data source file is a shortcut for quickly connecting to the original data that you use regularly. Data source file does not contain the actual data, and they only contain the necessary information to connect with the actual data. You can modify the top of the actual data such as creating calculated fields, changing default properties, adding groups, and so on.
Tableau Packaged Data Source	(.tdsx)	Tableau packaged data source is very similar to the tableau data source, but it has an addition of data along with the connection details.
Tableau Preferences	(.tps)	This file stores the color preferences, which is used among all the datasheets. It is also used to generate a customized look for the users.

Fig 1.7: Types of Tableau files

SELF-ASSESSMENT QUESTIONS – 2

3. What are the different Tableau files?
4. Tableau was founded by_____.

7. DATA: JOINING AND BLENDING

7.1 Data Connection with Data Sources

All readily available and widely used data sources can be connected to Tableau. It can connect to text files, Excel files, PDF files, and more. Using its ODBC connector, it can also establish connections to various databases. Tableau can connect to servers and web connectors.

Tableau's native connectors can connect to the following types of data sources:

- File Systems: Such as Microsoft Excel, CSV, etc.
- Cloud Systems: Such as Google big Query, Windows Azure, etc.
- Relational System: Such as Microsoft SQL Server, Oracle, DB2, etc.
- Other Sources: It uses ODBC.

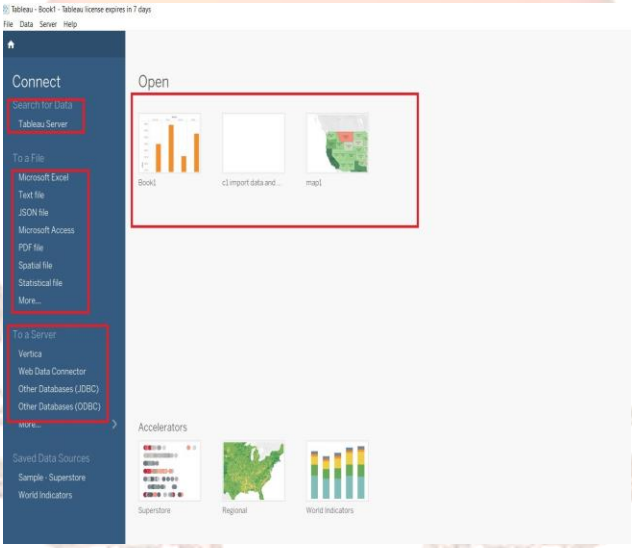

Fig 1.8: Different import options

7.2 How to Load and display File in Tableau?

The steps to load and display data are as follows:

- Step 1: Select dashboard from the new dashboard option.
- Step 2: Right-click on any open tab in the workbook.
- Step 3: Select New Dashboard from the menu.

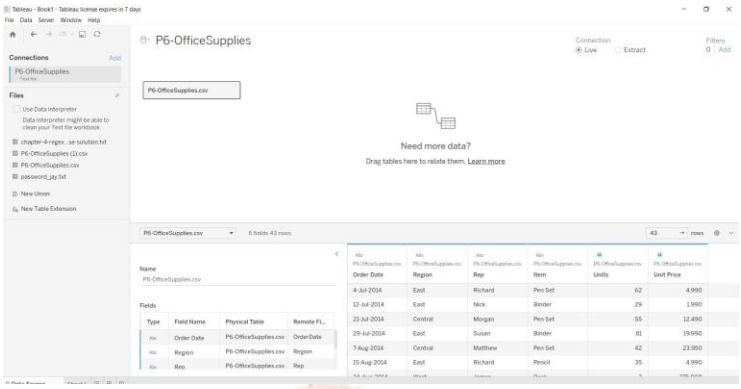

Fig 1.9: Different Views of data

7.3 Simple Demo Graph Representation

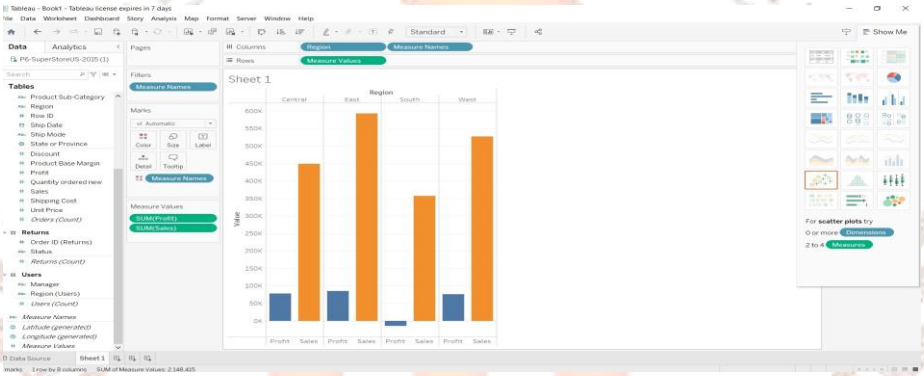

Fig 1.10: Bar Graph Comparison

Tableau Desktop

Data from several sources may be connected using Tableau Desktop to create dashboards, stories, and workbooks. You can publish the workbooks on the Tableau website and share all the insights with other users using the Tableau Desktop.

Without developing any code, a user of Tableau Desktop can do direct queries on the datasets. You only need to enter in the visualizations, such as a chart, table, graph, or map, and then write the columns you wish to include. Additionally, Tableau Desktop creates dashboards that mix numerous views from various data sources.

Tableau Public

This Tableau version was created with budget-conscious consumers in mind. The phrase "Public" indicates that the generated workbooks cannot be locally saved. They ought to be stored on Tableau's public cloud, which anyone can access and observe.

The files stored in the cloud have no privacy, therefore anyone can view and download the same information. For people who want to study Tableau and for those who wish to publish their data with the world, this version is optimal.

Tableau Online

Although its functionality is comparable to that of the tableau server, data is kept on servers hosted in the cloud that are managed by the Tableau group.

The data that is made available via Tableau Online can be stored indefinitely. Over 40 cloud-hosted data sources, including Hive, MySQL, Spark SQL, Amazon Aurora, and many more, are directly connected via Tableau Online.

The workbooks produced by Tableau Desktop must be published for Tableau Server and Tableau Online to function. Google Analytics and Salesforce.com are two web programs that Tableau Server and Tableau Online may access data from.

Tableau Server

The software is properly utilized to distribute workbooks and visualizations produced by the Tableau Desktop application around the company. You must publish your worksheet in Tableau Desktop before sharing dashboards on the Tableau Server. Only the authorized users will have access to the worksheet once it has been uploaded to the server.

Authorized users don't necessarily need to have Tableau Server installed on their computers. They merely need the login information in order to examine reports using a web browser. Tableau Server's high level of security is advantageous for efficient and speedy data exchange.

The organization's administrator has complete control over the server. Both the software and the hardware are maintained by the organization.

Tableau Reader

We may view the visualizations and workbooks made using Tableau Desktop or Tableau Public using the free utility Tableau Reader. Filtering the data is possible, but changes and editing are limited. Tableau Reader has no security because anyone may use it to read workbooks.

The recipient of the dashboards you build themselves must have Tableau Reader to read the file.

SELF-ASSESSMENT QUESTIONS – 3

5. Tableau is_____Software?
6. Tableau is used to find insight from data. True/False
7. Tableau integrates_____data sources.

8. GLOSSARY

- Visualization: finding insight from data
- Import data: Load the data into tableau for processing

9. TERMINAL QUESTIONS

1. **What are the five basic Features of any data visualization Software?**

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Speed of Analysis Since it does not call for a high level of programming knowledge, any user with access to data can begin utilizing it to extract value from the data.

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- All of the organization's published data sources may be managed in one place thanks to Tableau server. In one simple place, you can remove, modify permissions, add tags, and manage schedules. Extract refreshes may be easily scheduled and managed on the data server.

2. Explain Different types of files used in tableau.

After data processing, Tableau's output can be saved in a variety of formats and then delivered across several platforms. The several distinct extensions serve to distinguish the various types of different file categories. Their length varies depending on how they are produced and how they are used.

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Tableau Packaged workbook	(.twbx)	Tableau packaged workbook is a single zip file which contains a workbook along with any supporting local file data and background images. This is the best way to package your work for sharing with others who don't have access to the original data.
Tableau data Extract	(.hyper or .tde)	Tableau data extract is a local copy of the entire data set. It is used to share the data with others when you worked offline, and want to improve the performance.
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- Cloud Systems: Such as Google big Query, Windows Azure, etc.
- Relational System: Such as Microsoft SQL Server, Oracle, DB2, etc.
- Other Sources: It uses ODBC.

4. **Explain Tableau and Other Visualization tools.**

Data visualization uses visual components like graphs, charts, and maps to graphically portray quantitative information and data. Data visualization turns both huge and small data sets into graphics that are simple for people to comprehend and process. Data outliers, patterns, and trends can be easily understood with data visualization tools. The tools and technology for data visualization are essential in the realm of big data because they allow for the analysis of enormous amounts of data.

Top Data Visualization Tools

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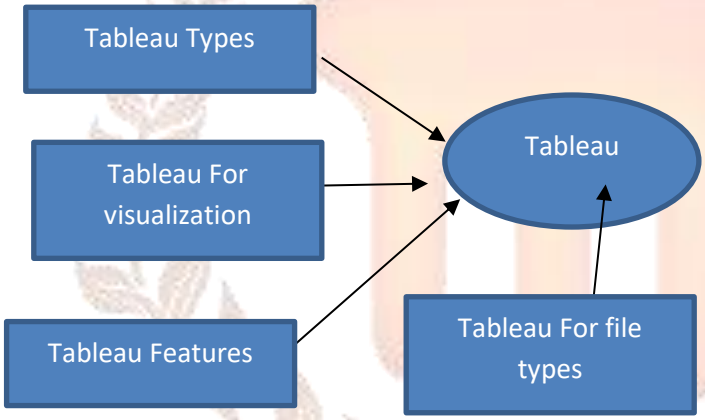
- **Plotly:** An interactive, open-source, and browser-based Python graphing toolkit is called plotly.py.Plotly.py is a high-level, declarative charting framework that is built on top of plotly.js. Plotly.js comes with more than 30 different chart kinds, including financial charts, scientific charts, 3D graphs, and more. Plotly is MIT licensed software. Plotly graphs can be seen in standalone HTML files, Jupyter notebooks, or Dash programmers' advice, dashboard creation, app integration, and feature requests, get in touch with us.
- **RAW:** RAW, also referred to as RawGraphs, operates on delimited data such as TSV or CSV files. It acts as a bridge between spreadsheets and data visualization. Despite being a web-based application, RawGraphs offers strong data protection and offers a variety of unconventional and traditional layouts.

10. SUMMARY

Tableau offers numerous appealing and distinctive features that make it a top tool for data visualization. You can quickly get the answers to significant queries because of its robust data finding and exploration application. Any data may be visualized using Tableau's drag and drop interface.

Explore various perspectives, and even seamlessly integrate numerous databases. It doesn't need any difficult scripting. Anyone who is familiar with the business issues can solve them by visualizing the pertinent facts. Sharing results with others is as simple as publishing to Tableau Server after analysis.

11. CONCEPT MAP



11. ANSWERS

Self-Assessment Questions

1. False
2. 2003
3. Workbooks, Bookmarks, Packaged Workbooks
4. Chris Stolte
5. data visualization
6. True
7. Multiple

12. REFERENCE

- Practical Tableau: 100 Tips, Tutorials, and Strategies from a Tableau Zen Master
- Tableau Your Data! Fast and Easy Visual Analysis with Tableau Software